

Sales Performance Analysis Using Python, SQL & Power BI

Subtitle : Data Storytelling & Statistical Validation

Project Information:

Prepared By:

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Internship:

ApexPlanet Software Pvt. Ltd

Role:

Data Analytics Intern

Tools Used:

Python, SQL, Power BI, Excel

Date:

July 2026



Business Problem & Objectives

Business Problem

The company wants to analyze its sales data to understand customer purchasing behavior, product performance, and sales trends. The goal is to identify the key factors affecting total sales and use data-driven insights to improve business performance and decision-making.

Project Objectives

- ▶ Analyze overall sales performance.
- ▶ Identify top-selling products and categories.
- ▶ Understand customer purchasing behavior based on age, gender, and city.
- ▶ Analyze sales trends over time.
- ▶ Compare sales performance across different cities.
- ▶ Generate business insights using Python, SQL, and Power BI.
- ▶ Provide recommendations to improve sales performance.

Why This Analysis Is Important

- ▶ Improve overall sales and revenue.
- ▶ Understand customer buying patterns.
- ▶ Identify the best-performing products and categories.
- ▶ Support better inventory planning.
- ▶ Help management make data-driven business decisions.
- ▶ Improve marketing strategies based on customer insights.

Dataset Overview

Dataset Features

- ▶ Order_ID
- ▶ Order_Date
- ▶ Customer_ID
- ▶ Customer_Name
- ▶ Age
- ▶ Gender
- ▶ City
- ▶ Product
- ▶ Category
- ▶ Quantity
- ▶ Unit_Price
- ▶ Total_Sales

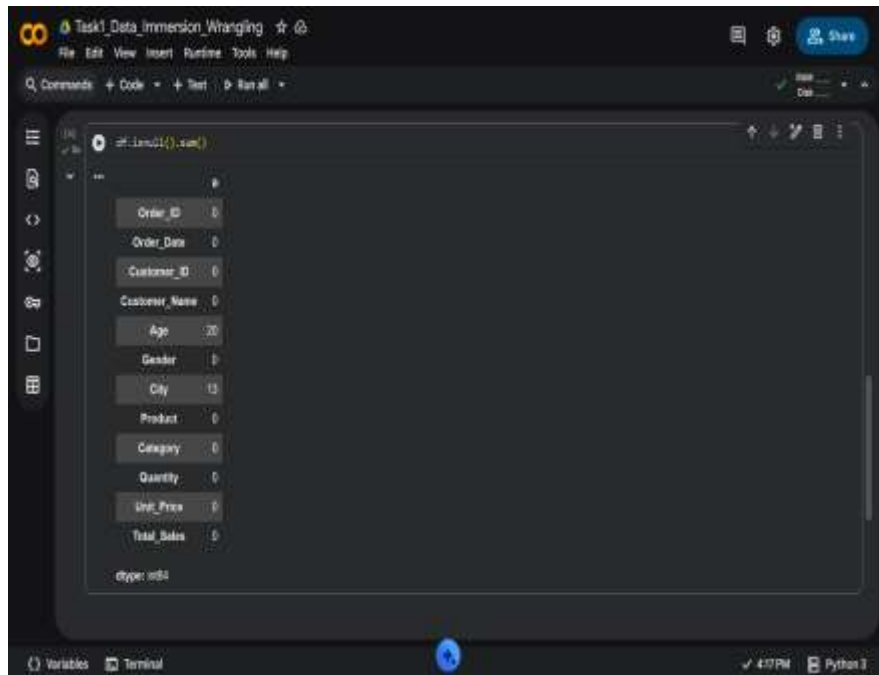
Attribute	Value
Dataset Name	ApexPlanet_Data Analytics_Dataset
Number of Records(Rows)	1001
Number of Features(Columns)	12
Data Type	Sales Transaction Data
Missing Values	No Missing Values
Duplicate Records	Checked and removed during preprocessing
Target Variable	Total_Sales

Data Cleaning & Preprocessing

Data Cleaning Process

- ✓ Checked missing values
- ✓ Removed unnecessary spaces
- ✓ Converted data types
- ✓ Standardized column names
- ✓ Verified duplicate records
- ✓ Prepared data for analysis

Before Cleaned Data



Order_ID	Order_Date	Customer_ID	Customer_Name	Age	Gender	City	Product	Category	Quantity	Unit_Price	Total_Sales
0	0	0	0	20	0	13	0	0	0	0	0

After Cleaned Data



```
# Drop Duplicates (inplace)
for col in df.select_dtypes(include=object):
    df[col] = df[col].astype(str).str.strip()

print("Data Cleaned Successfully")

# Save the cleaned data to a CSV file
df.to_csv("cleaned_data.csv", index=False)

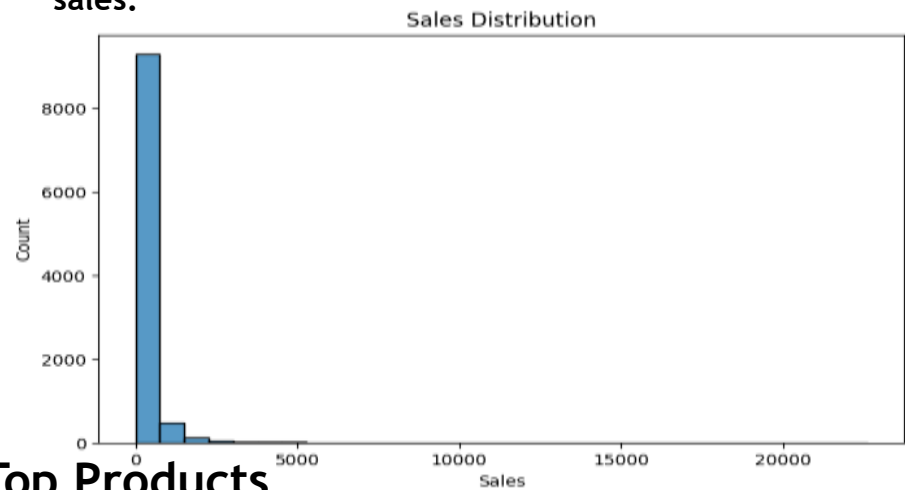
print("Cleaned Dataset Saved")

from google.colab import files
files.download("cleaned_data.csv")
```


Exploratory Data Analysis(EDA)

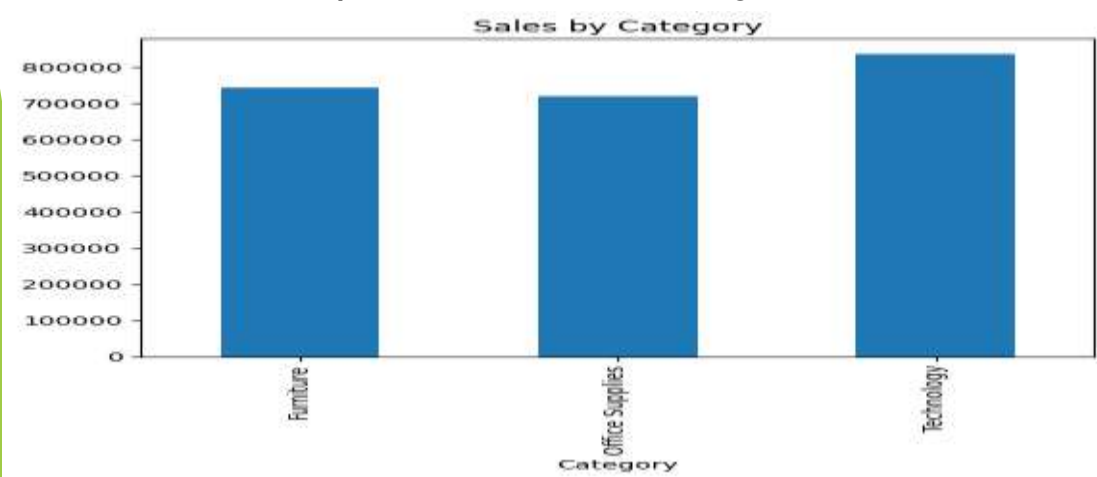
Sales Distribution(Histogram)

Most transactions have lower sales values, while a few high-value transactions contribute significantly to total sales.



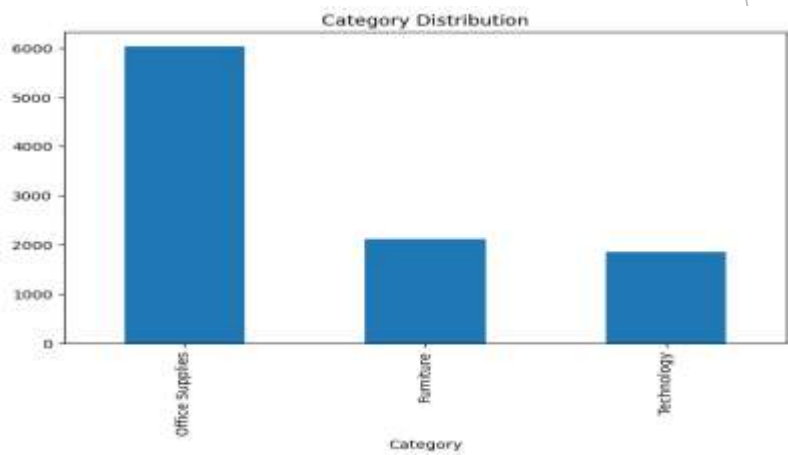
Top Products

A small number of products contribute the highest revenue.



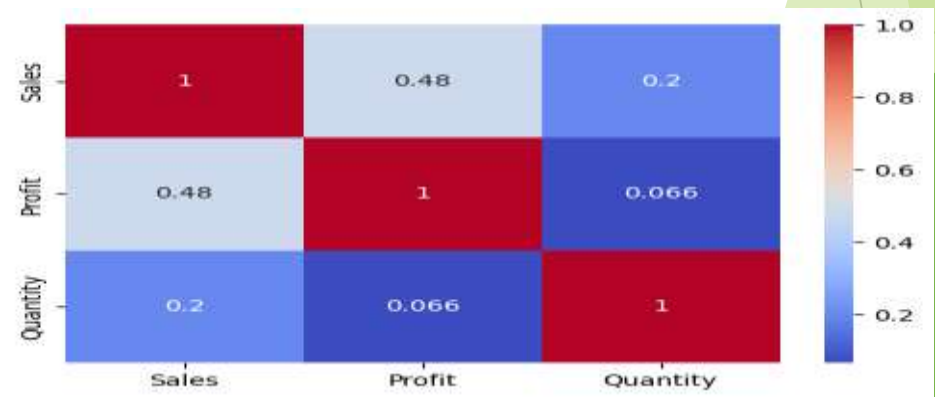
Category-wise Sales(Bar Chart)

Some product categories generate higher sales than others.



Correlations Heatmap

Positive correlations exist between some numerical variables, helping identify key business relationships.





SQL Analysis

Objective: SQL was used to query the sales database and answer key business questions efficiently. The analysis helped identify sales performance, product trends, and customer purchasing patterns

Business Questions Answered Using SQL

- ❖ ✓ What is the total sales generated by the business?
- ❖ ✓ What is the average sales per transaction?
- ❖ ✓ Which products generate the highest sales?
- ❖ ✓ Which product category contributes the highest revenue?

SQL Queries Used

1. Total Sales

Query: `SELECT SUM(Total_Sales) AS Total_Sales
FROM Sales;`

2. Average Sales

Query: `SELECT AVG(Total_Sales) AS Average_Sales
FROM Sales;`

SQL Queries Used

3.Top Selling Products

Query: **SELECT Product,
SUM(Total_Sales) AS Total_Sales FROM
Sales GROUP BY Product
ORDER BY Total_Sales
DESC;**

4.Category-wise Sales

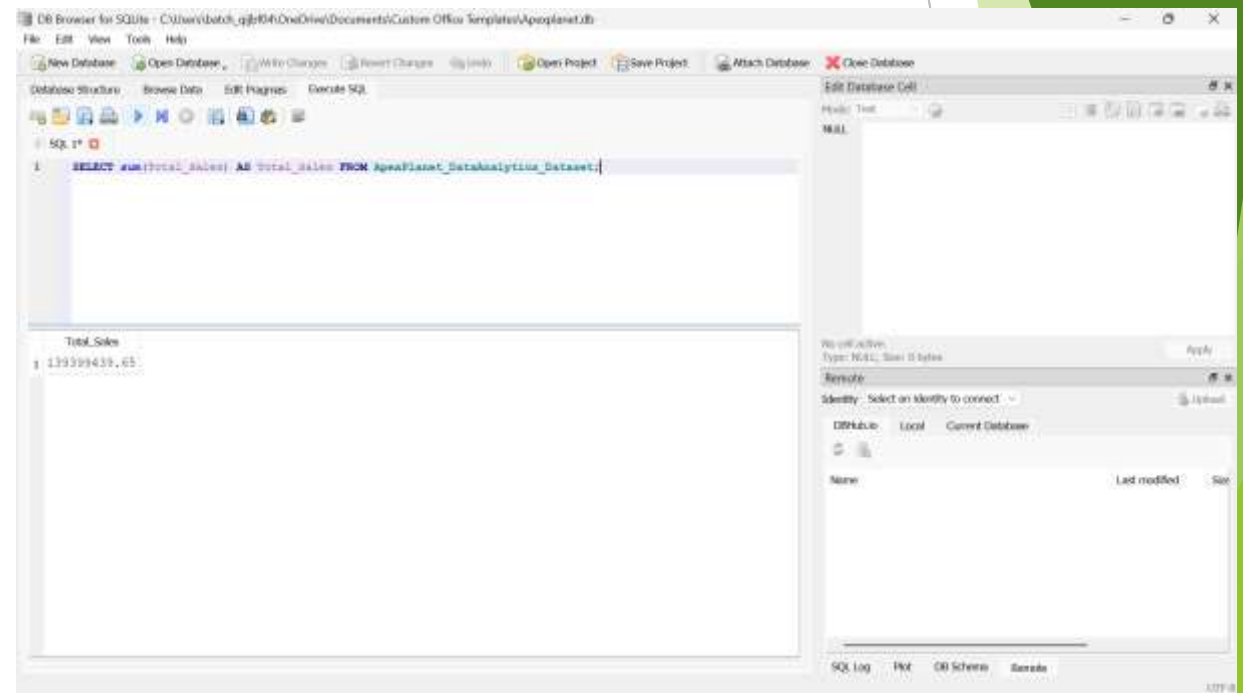
Query: **SELECT Category,
SUM(Total_Sales) AS Total_Sales FROM
Sales GROUP BY Category
ORDER BY Total_Sales DESC;**

Key Insights

- SQL enabled quick business analysis.
- Identified top-performing products.
- Summarized sales information efficiently.

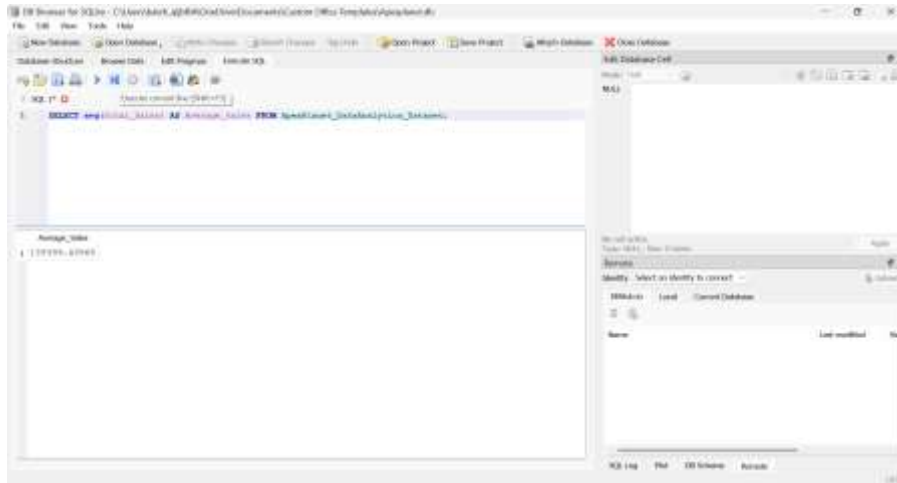
1.Total Sales

Query: **SELECT SUM(Total_Sales) AS Total_Sales FROM
Sales;**



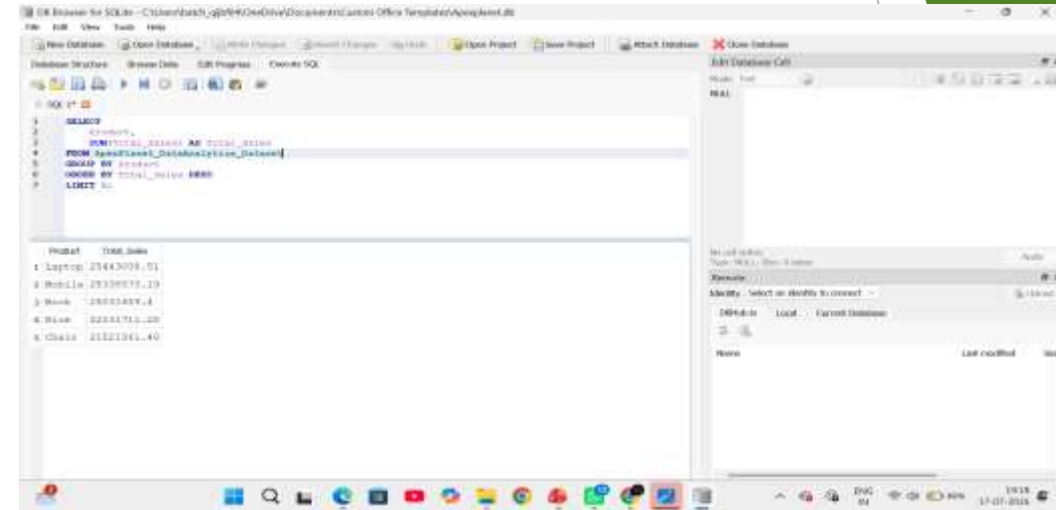
2. Average Sales

Query: SELECT AVG(Total_Sales) AS Average_Sales FROM Sales;



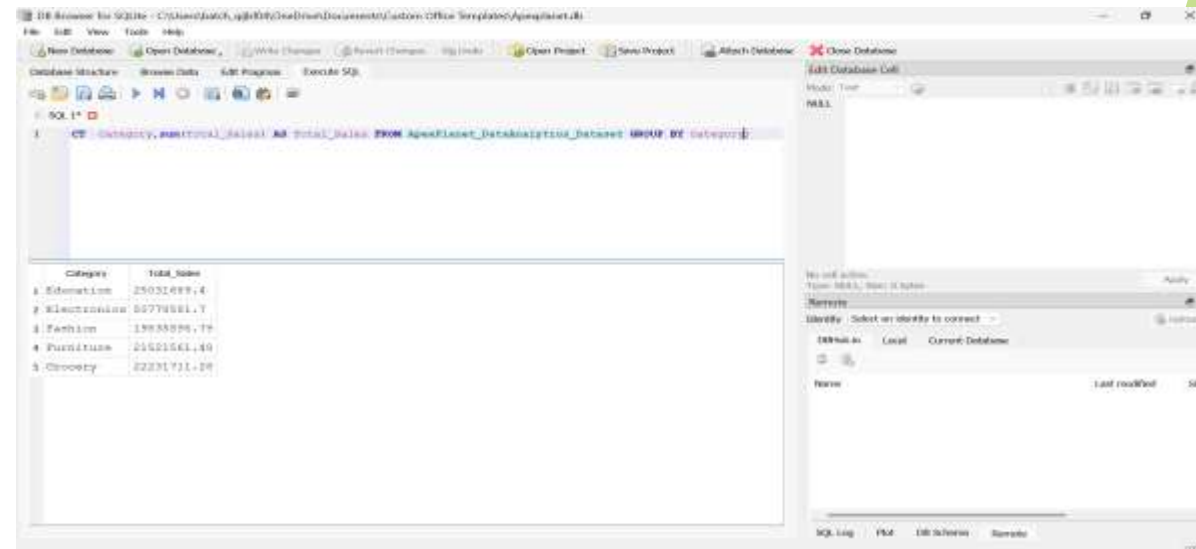
3. Top Selling Products

Query: SELECT Product, SUM(Total_Sales) AS Total_Sales FROM Sales GROUP BY Product ORDER BY Total_Sales



4. Category-wise Sales

Query: SELECT Category, SUM(Total_Sales) AS Total_Sales FROM Sales GROUP BY Category ORDER BY Total_Sales DESC;



Power BI Dashboard

KPIs

- ✓ Total Sales
- ✓ Total Orders
- ✓ Average Sales
- ✓ Total Quantity Sold

Interactive Filters

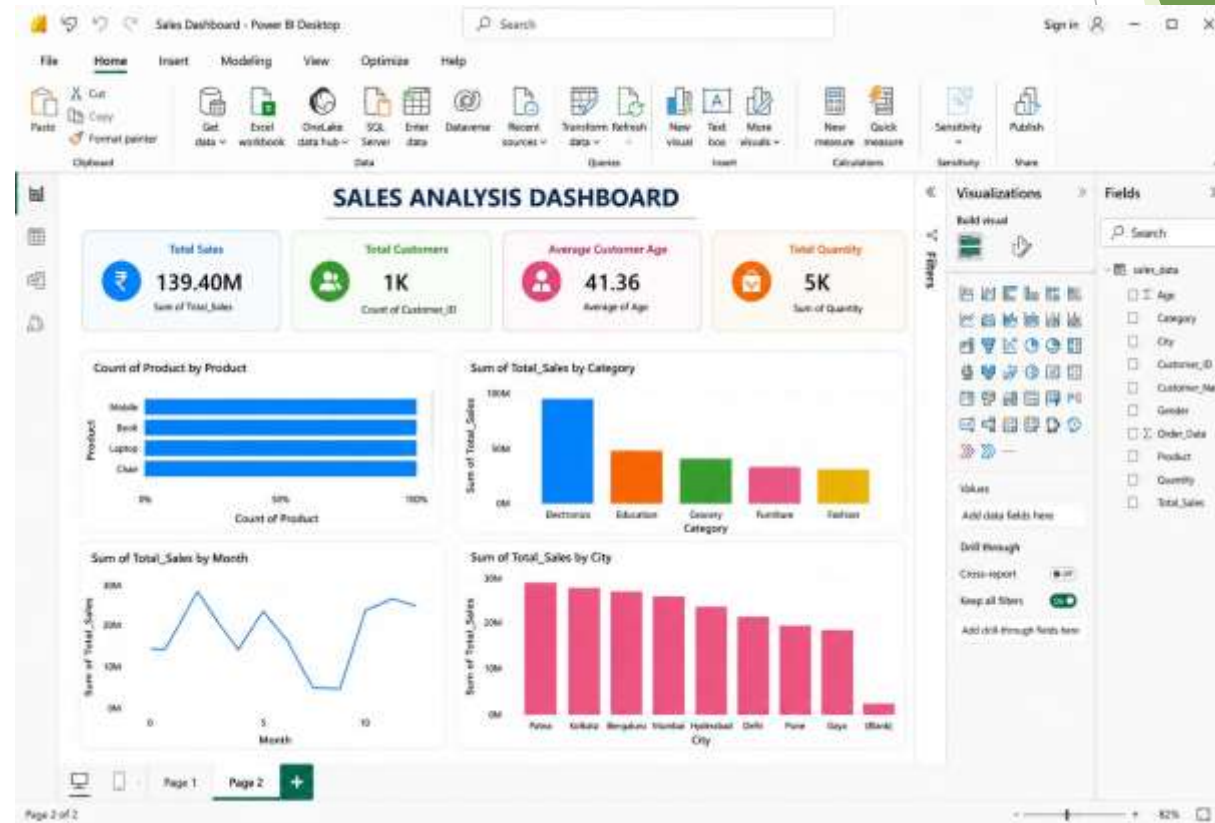
- ✓ Category
- ✓ Product
- ✓ City
- ✓ Gender

Visualizations

- ✓ Bar Chart
- ✓ Histogram
- ✓ Heatmap
- ✓ KPI Cards

Key Benefit

- Interactive dashboard for business decision-making.



Business Insights

Key Business Insights

- ▶ Electronics generated the highest total sales, making it the best-performing product category.
- ▶ Education ranked second in total sales, followed by Grocery, Furniture, and Fashion.
- ▶ Laptop is the top-selling product, followed closely by Mobile, indicating strong demand for electronic products.
- ▶ Patna recorded the highest total sales among all cities, followed by Kolkata, Bengaluru, Mumbai, and Hyderabad.
- ▶ The customer base is well balanced, with 511 male customers and 489 female customers.
- ▶ Sales transactions cover multiple categories and cities, helping identify regional and product-level business opportunities.
- ▶ The analysis highlights that electronic products contribute significantly to overall business revenue.

Statistical Validation

Hypothesis

- ▶ Null Hypothesis (H_0): There is no significant relationship between Quantity Sold and Total Sales.
- ▶ Alternative Hypothesis (H_1): There is a significant relationship between Quantity Sold and Total Sales.

Statistical Test Used

- ▶ Pearson Correlation Test

Results

- ▶ Correlation Coefficient (r): 0.647
- ▶ p-value: 1.71×10^{-119} (approximately 0.000)

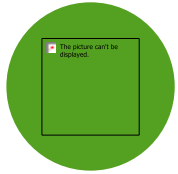
Decision

- ▶ Since p-value < 0.05 , Reject the Null Hypothesis (H_0).

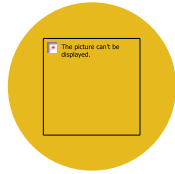
Conclusion

- ▶ There is a statistically significant positive relationship between Quantity Sold and Total Sales. This means that, in general, as the quantity of products sold increases, the total sales value also increases.

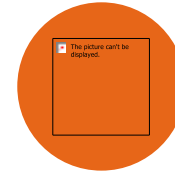
Business Recommendations



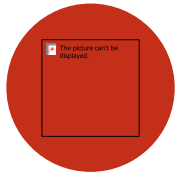
Recommendations



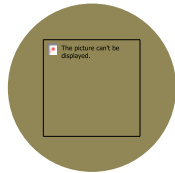
✓ Maintain sufficient inventory for Electronics products, especially Laptops and Mobiles, to meet customer demand and avoid stock shortages.



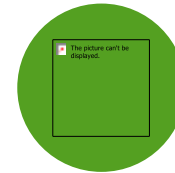
✓ Increase marketing efforts in high-performing cities such as Patna, while creating targeted campaigns to improve sales in lower-performing cities.



✓ Develop promotional offers for lower-performing product categories to increase customer interest and boost sales.



✓ Analyze customer purchasing behavior regularly to identify buying trends and personalize marketing strategies.



✓ Monitor sales performance using the Power BI dashboard to support timely, data-driven business decisions.



✓ Use sales trends to improve inventory planning, ensuring that high-demand products are always available.

Conclusion

Successfully cleaned and preprocessed the ApexPlanet_DataAnalytics_Dataset-1 to ensure data quality and consistency.

Performed Exploratory Data Analysis (EDA) using Python to identify sales patterns, customer behavior, and product performance.

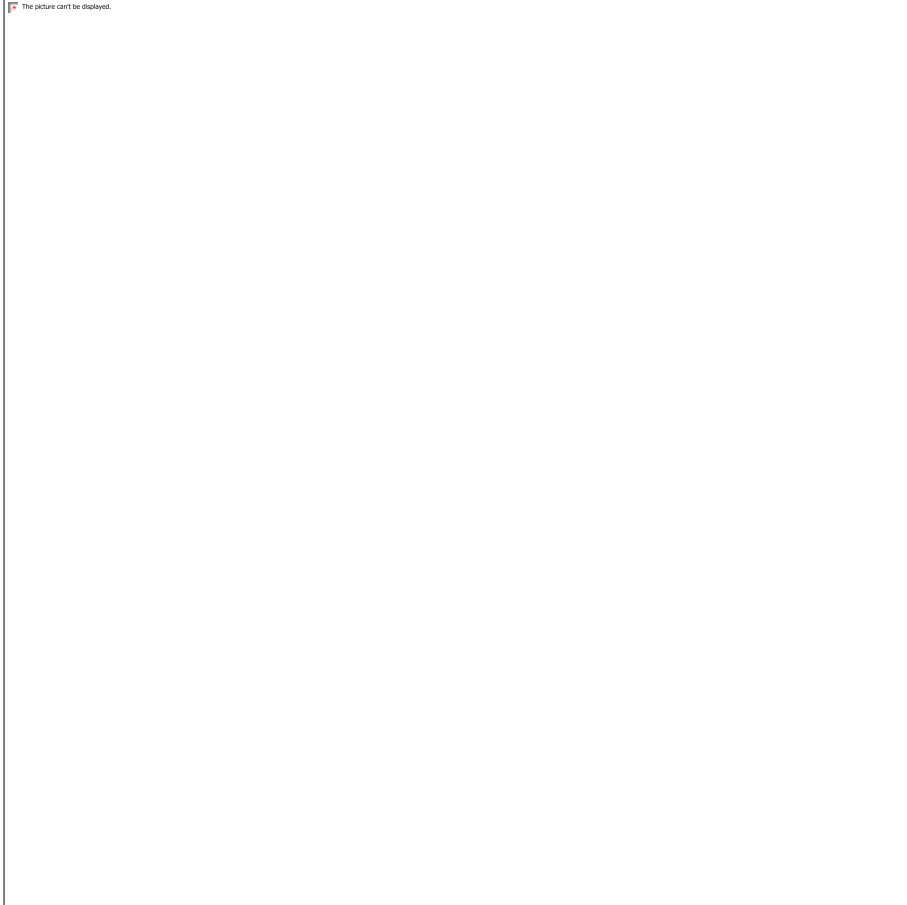
Used SQL to answer key business questions, including total sales, average sales, top-selling products, and category-wise sales.

Developed an interactive Power BI dashboard to visualize KPIs, sales trends, and business performance.

Generated valuable business insights, including the highest-performing product categories, top-selling products, and city-wise sales performance.

Validated the findings using Pearson Correlation Analysis, confirming a statistically significant positive relationship between Quantity Sold and Total Sales.

Proposed practical business recommendations to improve sales performance, inventory management, and decision-making.



Thank you